JAYANT SONAWANE

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PROFILE

Data & AI hands-on leader with 10+ years of experience turning complex data challenges into measurable business outcomes. From data platorms to MLOps and AI-powered analytics, I built platforms that deliver-10x faster insights, ~80% reduction in infrastructure costs, and 90% fewer critical incidents. Proven at building from scratch (0 to 1) and scaling at pace, consistently aligning technical execution with business strategy in regulated environments.

EDUCATION

Texas A&M University-Kingsville **May 2017**

Master of Science, Computer Science, **GPA 4.0**

Pune Institute of Computer Technology **May 2013**

Bachelor of Engineering, Information Technology

TECHNICAL SKILLS

Programming: Python, SQL, Scala
Data Skills: Pandas, NumPy, ETL, Spark, Airflow, Kafka, Iceberg, dbt
Data & AI: Data Strategy, Data Architecture, Data Governance, Ethical AI, Agentic AI, AI-Assisted Development
AI Tools/Frameworks: LangGraph, LangChain, LLM, RAG, Vector DBs, Cursor, Claude Code
Cloud Platforms: AWS, GCP, Oracle, Azure
Data Warehousing: Redshift, Snowflake, Netezza
BI & Visualization: Power BI, Tableau, Superset
DevOps & Infra: Docker, Terraform, Terragrunt, AWS ECS, Kubernetes, Github Actions, CI/CD
Web Technologies: HTML5, CSS3, JavaScript, React, Flask

PROFESSIONAL EXPERIENCE

Senior Manager, Data Engg & Analytics, TPConnects (Airline Retailing), Dubai, UAE **Jun 2024 — Present**

- Defined and executed an enterprise data strategy and multi-geo data architecture for TPC.
- Architected and implemented an AI powered transactional Data Lakehouse (Single source of truth) on AWS from scratch cutting infra cost by 80% and P1 incidents by 95%.
- Designed and implemented a real-time airline Look-to-Book (L2B) analytics engine which helped to lower operational costs by ~30%.
- Designed and implemented conversational AI-powered Data Analytics Assistant using AWS Bedrock, LangGraph and RAG for democratizing data access and enabling self-serve analytics.
- Developed financial reporting and analytics system, reducing manual overhead by over ~80%.
- Delivered an AI powered data observability agent, monitoring pipelines and data quality across data platform.
- Spearheaded the development of Agency Travel Insights platform, embedded within business application, unifying data insights within operational tools and driving agency productivity.
- Led a multi-geo data team, driving cross-functional data strategy, execution, and governance across TPC.
- Stack: Python, SQL, Spark, Apache Iceberg, Airflow, AI/LLM, Terraform/Terragrunt, AWS (Athena, S3, Glue, Kinesis, Bedrock), RAG, LangGraph, Cursor, Docker, CI/CD, Superset, PowerBI, Tableau.

Data Engineering Manager, Tarabut (Open Banking FinTech), Dubai, UAE **May 2022 — May 2024**

- Delivered a multi cloud enterprise data platform supporting analytics and data products across 4 regulated jurisdictions.
- Designed and implemented reusable ETL orchestration framework, improving delivery speed and consistency across pipelines by >50%.
- Led implementation of MLOps pipelines for ML powered bank transaction enrichment service, achieving >99% pipeline reliability while reducing manual overhead by ~40%.
- Drove adoption of data contracts architecture, resulting in highly reliable data pipelines and near-zero production failures.

- Automated AWS infra deployment with Terraform and CI/CD (IaC) —standardized environments and reduced deployment time by 70%.
- Scaled the data function (0->1), partnered with 5 cross-functional squads to deliver business critical analytics use cases.
- Stack: Python, SQL, Kafka,MLOps, NLP, Airflow, Apache Iceberg, AWS (Athena, S3, EC2, RDS, Glue, Lambda), Terraform, Docker, Power BI, Superset.

Lead Data Engineer, Rain Financial (Crypto Trading Platform FinTech), Dubai, UAE **Jun 2021 — May 2022**

- Built an AWS Lakehouse from scratch for RAIN —ingesting tick-level/transactional data to power analytics and data science use cases.
- Designed and built a low-latency crypto pricing ingestion service, enabling real-time pricing analytics and business decision-making.
- Created a modular data pipeline framework (Python, SQL, Airflow) that reduced time for creating production-ready data pipelines by ~40%.
- Built automated data quality testing framework using Python and Great Expectations.
- Improved query performance by ~80%, accelerating decision-making and reducing wait times for critical insights.
- Led a team of 2 data engineers and partnered with cross-functional business stakeholders to deliver impactful analytics use cases.
- Stack: Python, SQL, Airflow, Apache Arrow, Pandas, TimescaleDB, AWS (Athena, Redshift, S3, EMR, RDS), CDK, ECS, Docker, CI/CD.

Senior Data Engineer, CAFU (Vehicle Services Platform), Dubai, UAE **Oct 2019 — May 2021**

- Architected a data platform on AWS using modern data stack for last-mile logistics; built batch/streaming pipelines (Kafka/Spark/Airflow) processing 100M+ events/month.
- Delivered ML pipelines with Airflow and SageMaker for churn and ETA prediction use cases—improved SLA adherence and route efficiency.
- Drove end-to-end ETL/ELT and data infra deployment modernization (Airflow,AWS CDK, GitLab)—cut release cycles by 50%.
- Partnered with BI, Product and Operations teams to deliver near real-time dashboards for dispatch and fleet.
- Stack: Python, SQL, Pandas, Parquet, AWS (Athena, Redshift, S3, CDK, ECS, EMR, SageMaker), PySpark, Airflow, Kafka, Docker, PowerBI.

Senior Data Engineer, PropertyFinder (Real Estate Portal), Dubai, UAE **Feb 2019 — Oct 2019**

- Served as key member for development of data lake architecture on AWS for storing TBs of data.
- Led end to end migration of Hadoop based data lake to AWS and Snowflake---reducing compute costs by 30%.
- Implemented data pipelines and ETL processes using Python 3, Airflow, AWS and Fivetran.
- Architected and delivered data APIs that fueled BI dashboards and actionable product insights for real estate agents..
- Migrated codebase from Python 2 to Python 3, provisioned infra with Terraform, and containerized services using docker and ECS—standardized delivery.

Software Engineer, Qualcomm (Cellular and Semiconductor), San Diego, USA **Jan 2017 — Feb 2019**

- Developed scalable data pipelines and ETL workflows using Apache NiFi, Python, PostgreSQL, and AWS.
- Migrated Hadoop based data environment to AWS data lake and Redshift data warehouse.
- Designed and built high-performance APIs with Python, Pandas, PostgreSQL, and FastAPI to accelerate access to network and satellite data.
- Utilized big data frameworks like Hadoop, Spark, Hive, and MapReduce for data processing and transformations.
- Delivered new features for internal project management applications using the .NET framework.

Engineer-Summer Intern, Qualcomm, San Diego, USA **Jun 2016 — Aug 2016**

- Performed large-scale data analysis using Netezza SQL.
- Built ETLs using Python and Pandas.
- Automated JIRA workflows using Python and Flask.

Programmer Analyst, Cognizant Technology Solutions (Tech Services), Pune, India **Jan 2014 — Jul 2015**

- Partnered with Credit Suisse investment bank to enhance trading platform performance and reliability.
- Optimized the trade capture system in C++, improving performance and delivering a more cost-efficient solution.
- Migrated legacy C++ codebase to newer product versions, improving maintainability and stability.
- Diagnosed and resolved critical production defects, strengthening system robustness.
- Automated test suites using Python (Pytest), increasing test coverage and efficiency.
- Provided production support for mission-critical systems.